🔐 Project Title: Secure Login System with Validation (Java OOP)

📌 Project Description

This project is a simple cybersecurity-based application developed using Java and Object-Oriented Programming (OOP). The system simulates a basic secure login environment where users can register and log in while following essential security rules.

The main goal of this project is to demonstrate core cybersecurity concepts such as authentication, password strength validation, and protection against unauthorized access in a simplified and understandable way.

🎯 Objectives

- To implement a secure user login system
- To validate user credentials (email and password)
- To prevent weak passwords
- To restrict multiple failed login attempts (basic brute-force protection)
- To apply OOP concepts like classes, objects, and modular design

---

⚙ System Features

1. User Registration

- User enters username, email, and password
- System validates:
  - Email must contain “@” and a domain (e.g., .com)
  - Password must meet basic strength rules

2. Password Strength Checker

- Checks:
  - Minimum length (e.g., 8 characters)
  - Contains uppercase and lowercase letters
  - Contains numbers
  - Optional: special characters
- Output: Weak / Medium / Strong

3. Login System (Authentication)

- User enters username and password
- System verifies credentials

4. Account Lock Mechanism

- Maximum 3 incorrect login attempts allowed
- After 3 failed attempts, account is temporarily locked

□ OOP Design (Classes Used)

1. User Class

Stores user information:

- Username
- Email
- Password
- Login attempts

2. Validator Class

Handles validation logic:

- Email validation
- Password strength checking

3. LoginSystem Class

Manages system operations:

- Register user
- Authenticate login
- Track login attempts

4. MainApp Class

- Runs the application
- Handles user interaction (console or GUI)

🛡 Cybersecurity Concepts Applied

- Authentication (verifying user identity)
- Password security (strong password enforcement)
- Brute-force attack prevention (login attempt limit)
- Input validation (email and password checks)

💻 Technologies Used

- Java
- OOP Concepts (Classes, Objects, Encapsulation)
- Optional: Java Swing (for GUI)

🚀 Expected Outcome

The system will allow a user to securely register and log in while enforcing basic security rules. It will demonstrate how simple security mechanisms can protect user accounts from weak passwords and repeated unauthorized access attempts.

📌 Future Enhancements (Optional)

- Add graphical user interface (GUI)
- Store multiple users
- Encrypt passwords (basic hashing)
- Add password visibility toggle

✅ Conclusion

This project provides a beginner-friendly implementation of a secure login system while applying core cybersecurity principles. It is simple, practical, and suitable for understanding both OOP and basic security mechanisms.

Complete Flow (Step-by-Step, Simple)

Think of your system in **2 phases**:

□ Phase 1: Registration (First Time Only)

This is where validation happens.

1. User enters:
   - username
   - email
   - password
2. System checks:
   - Email valid?
   - Password strong enough?

☞ If password is **weak**:

- System says: “Weak password, please enter again”
- User must re-enter password until it becomes acceptable

☞ Once everything is valid:

- Data is **saved**
- Registration complete ✅

⬤ Phase 2: Login

Now validation is NOT repeated here.

1. User enters:
   - username
   - password
2. System checks:
   - Does it match stored data?

☞ If correct:

- Login successful ✅

☞ If wrong:

- Attempts +1
- After 3 attempts → account locked ❌

OOP Structure

1. User Class

- Stores username, email, password
- Tracks login attempts

2. Validator Class

- Checks email format
- Checks password strength

3. LoginSystem Class

- Handles registration
- Handles login verification
- Manages account lock

4. MainApp Class

- Controls program execution
- Handles user interaction

🛡 Cybersecurity Concepts Used

- Authentication (login system)
- Password security (strong password enforcement)
- Brute-force protection (limited login attempts)
- Input validation

💻 Technologies

- Java
- OOP (Classes, Objects, Encapsulation)
- Optional: Java Swing (for GUI)